

Monisha A

📍 Vellore, Tamil Nadu, India | 📞 +91 9344515479 | ✉ monishaanand553@gmail.com
🌐 LinkedIn: linkedin.com/in/monishaanand175 ([link](#)) | 📄 GitHub: github.com/Monisha175 ([link](#))

Professional Summary

Software Developer with hands-on experience in full-stack application development, AI-powered automation, backend development, and cloud-native technologies. Experienced in building REST APIs using Spring Boot, developing real-time applications, integrating AI workflows, and improving Java application performance in cloud environments. Skilled in Python, Java, React, Docker, Kubernetes, Linux, Git, and RESTful Microservices, with internship experience at Nokia and SmartBridge. Passionate about developing scalable software solutions and intelligent automation systems.

Education

Vellore Institute of Technology (VIT), Vellore

Integrated M.Tech – Computer Science & Engineering (Virtusa Collaboration)

2021 - 2026

CGPA: 8.97

Seventh Day Adventist Matric Hr Sec School

Higher Secondary Education

Secondary School Education

2019 - 2021

Marks: 90.48%

Marks: 90.8%

Work Experience

Nokia — Performance Development Intern

August 2025 – May 2026

- Worked on **performance optimization** of Java applications running in cloud-native environments by analyzing JVM behavior, garbage collection, heap memory, and multithreading.
- Supported **testing and debugging** activities by identifying performance bottlenecks using JPS, JStat, Java Flight Recorder (JFR), heap dump analysis, and application logs.
- Collaborated with development teams to improve **system integration** between monitoring tools and Java-based services using Grafana, Prometheus, Kafka, and OpenTSDB.
- Automated performance monitoring tasks using Bash scripts on Linux to improve operational efficiency.
- Analyzed and optimized multithreaded application behavior including deadlock detection, thread pool tuning, and concurrency bottleneck resolution in Java services, improving throughput by approximately 28%.
- Assisted in deploying and monitoring containerized applications using Docker and Kubernetes.
- Developed automated runbooks and technical documentation for JVM tuning procedures (heap sizing, GC policy selection, JIT optimization), enabling repeatable observability and performance benchmarking workflows.
- Worked in an **Agile development environment**, participating in sprint planning, daily stand-ups, and collaborative code reviews.
Technologies: Java, Docker, Kubernetes, Linux, Grafana, Prometheus, Kafka, OpenTSDB, Bash

SmartBridge — Generative AI Intern

June 2024 – July 2024

- Developed AI-powered applications using **Large Language Models (LLMs)**, **Natural Language Processing (NLP)**, and **deep learning**.
- Fine-tuned Generative AI models and optimized prompt engineering for AI-driven automation.
- Built AI workflows by integrating machine learning models into intelligent automation solutions.
- Performed testing and debugging of AI models to improve prediction accuracy and response quality.
- Gained hands-on experience with **Google Cloud Vertex AI** while completing multiple **Google Cloud Skill Badges** as part of the internship training.
- Collaborated with mentors in an iterative development environment to improve AI model performance and deployment.
Technologies: Python, Vertex AI, Google Cloud Platform, LLMs, NLP, TensorFlow, PyTorch, Prompt Engineering

BrightGeeks — AWS Cloud Intern

Jan 2024 – Mar 2024

- Completed hands-on training aligned with the AWS Certified Cloud Practitioner curriculum.
- Gained hands-on experience with AWS services including Amazon EC2, Amazon S3, Amazon RDS, IAM, VPC, CloudWatch, and AWS Lambda.
- Gained knowledge of cloud computing concepts, AWS global infrastructure, security, pricing, and shared responsibility model.
- Practiced deploying and managing cloud resources through the AWS Management Console.
- Developed a foundational understanding of cloud architecture, networking, storage, monitoring, and security best practices.
Technologies: AWS EC2, S3, IAM, VPC, RDS, CloudWatch, Lambda

Technical Skills

- **Languages:** Java, Python, C, SQL, HTML5, CSS3, Bash, JavaScript

- **Frontend:** React, Angular, Vue.js, WebSockets
- **Backend:** SpringBoot, Django, Flask, Node.js, REST APIs, Microservices
- **Cloud Platforms:** Google Cloud Platform (Vertex AI) Amazon Web Services (EC2, S3, IAM, VPC, RDS, Lambda)
- **DevOps:** Docker, Kubernetes, Helm, Linux, Jenkins, Git, Maven, CI/CD Pipelines
- **Monitoring & Networking:** Grafana, Prometheus, Apache Kafka, OpenTSDB, Netconf
- **AI & Machine Learning:** TensorFlow, PyTorch, Prompt Engineering, Generative AI, Large Language Models (LLMs), Natural Language Processing(NLP), AWS (Course Completion), GCP Vertex AI
- **Soft Skills:** Analytical Thinking, Cross-functional Collaboration, Technical Documentation, Problem-Solving

Projects

Autonomous Car Driving using CNN

2025

Python | TensorFlow | Deep Learning

GitHub Repository: <https://github.com/Monisha175/Self-Driving-cars>

- Developed a deep learning model using **Convolutional Neural Networks (CNNs)** for autonomous vehicle object detection and lane recognition.
- Preprocessed and augmented image datasets to improve model training accuracy and generalization.
- Designed and trained CNN architectures using Python and TensorFlow for real-time image classification.
- Evaluated model performance using accuracy, precision, and recall metrics to optimize prediction quality.
- Improved object detection accuracy, enabling faster and more reliable driving decision-making.
- Performed model testing, debugging, and performance optimization to enhance prediction efficiency.

Technologies: Python, TensorFlow, CNN, Deep Learning, Computer Vision, Machine Learning

AI-Powered Optimized Spraying Techniques for Pest Disease Control

2025

Python | Machine Learning | Data Analysis

GitHub Repository: github.com/Monisha175/AI-Spraying-System

- Designed and developed an AI-powered crop disease detection and smart spraying recommendation system.
- Built machine learning models to identify crop diseases from agricultural datasets.
- Applied data preprocessing, feature engineering, and model evaluation techniques to improve prediction accuracy.
- Generated intelligent spraying recommendations to minimize excessive pesticide usage.
- Reduced chemical consumption by **30%** while maintaining crop health and productivity.
- Improved automation and decision-making for precision agriculture applications.

Technologies: Python, Machine Learning, Data Analysis, Artificial Intelligence

Real-Time Chat Application using Spring Boot & WebSockets

2024

Java | Spring Boot | REST APIs | WebSockets

GitHub Repository: <https://github.com/Monisha175/Realtime-Chat-App>

- Developed a full-stack real-time chat application using **Spring Boot, REST APIs, and WebSockets**.
- Developed both backend REST APIs using Spring Boot and frontend components to enable seamless real-time communication between users.
- Designed and implemented RESTful APIs for user authentication, messaging, and chat management.
- Implemented WebSocket-based bidirectional communication for instant message delivery.
- Integrated backend services to support seamless real-time communication between multiple users.
- Performed testing and debugging to ensure reliable message transmission and application stability.
- Applied event-driven architecture concepts to improve application responsiveness.
- Used Maven for dependency management and Git for version control during development.

Technologies: Java, Spring Boot, REST APIs, WebSockets, Maven, Git

Certifications

- Nokia Performance Development Internship Completion Certificate – [View Certificate](#)
- Google Cloud Skills Boost Profile (*Earned during SmartBridge GenAI Internship*) – [View Profile](#)
- Generative AI Applications using Vertex AI (Google Cloud) – [View Certificate](#)
- AWS Cloud Practitioner Training – BrightGeeks – [View Certificate](#)
- Merit Scholarship Certificate – [View Certificate](#)

Extracurricular Activities

- **Red Cross Club Member** – Organized blood donation drives and health awareness campaigns.

Languages

- English (Proficient) • Tamil (Native) • Hindi (Intermediate)